# ALCHEMUS

## Advanced Drug Analysis Platform

Research Owner Wallet

A8ov5UqDdHWUAr8J81BCdUdoewNAPaGAtyH7TvQTYPoM

Date: 4/30/2025
Compounds: Bismuth Subsalicylate, Caffeine



## New Compound Analysis

Name: Bismuth Cafesalate
Formula: $C_{16}H_{12}BiN_4O_9$
Weight: 637.25
Drug-likeness: 0.65

Activity:
Antimicrobial and CNS Stimulant

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## Target Analysis

Prostaglandin G/H synthase 2
Binding Affinity: -8.1
Caffeine is known to inhibit phosphodiesterases, potentially affecting inflammatory pathways via prostaglandin synthesis inhibition.

Adenosine A2A receptor
Binding Affinity: -7.5
Caffeine's antagonistic action on adenosine receptors could alleviate fatigue, but increase chances of anxiety and insomnia.

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## Toxicity Profile

Risk Level: Moderate
Affected Systems: Kidneys, Liver, Gastrointestinal tract

Mitigation Suggestions:
- Limit cumulative dose
- Monitor renal function
- Provide adequate hydration

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## Pharmacokinetics

Absorption:
Rapid absorption in the gastrointestinal tract

Distribution:
Wide distribution with passage across the blood-brain barrier

Metabolism:
Hepatic, primarily via CYP1A2

Excretion:
Renal elimination predominantly as inactive metabolites

Half-life: 4-6 hours for caffeine component

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## Detailed Analysis

Bismuth Cafesalate is a novel compound hypothesized to have both antimicrobial and CNS stimulant properties by combining key elements of Bismuth Subsalicylate's antimicrobial capabilities and caffeine's stimulating effects. The compound was synthetically modeled to bring together bismuth's stabilized ionic attributes and caffeine's alkaloid structure, creating a fusion that retains essential activity across both components. Despite its potential, there are inherent toxicity risks, particularly affecting renal and gastrointestinal systems due to cumulative doses. A balanced absorption and metabolism profile suggests transient activation patterns with central nervous system passage. Further research and trial design have positioned this compound as a potential combined therapeutic, promising practical applications in managing mild infections while simultaneously mitigating fatigue. Carefully modeled through computational and lab-guided measures, this compound could revolutionize specific therapeutic mechanisms.

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